

Leah Uriarte

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EDUCATION

Harvey Mudd College

Expected May 2028

B.S. Computer Science - 3.8 GPA, Dean's List

INTERNSHIP EXPERIENCE

Incoming Software Developer Intern, [Google](#)

May 2026 - August 2026

- Core Infrastructure

Associate Software Developer Intern, [Google](#)

May 2025 - August 2025

- Architected book classification system with large language models, processing 23 million books at 93% genre accuracy
- Collaborated with partner to build bulk inference pipeline with Jax on Apache Beam in Python, enabling distributed inference across millions of records
- Achieved 95% precision, recall for BISAC classification and implemented semantic similarity evals with embeddings

RESEARCH EXPERIENCE

Researcher, Quantum Qasars Lab at Harvey Mudd College

Aug 2025 - Present

- Conducted outdoor telescope tests measuring bit error rate, signal strength, and alignment accuracy
- Programmed FPGAs and Raspberry Pis to control LED transmissions for quantum verification experiments

Researcher, Analog Circuit Engineering Lab at Harvey Mudd College

Sept 2024 - May 2025

- Built end-to-end data processing pipeline in Python with Pandas and NumPy, implementing complex database merging, temporal analysis, and statistical validation across multi-semester datasets
- Developed framework implementing inter-rater reliability testing and statistical comparisons for construct validation

Lead Researcher, Aspiring Scholars Directed Research Program

Sept 2022 - Feb 2024

- Led team of 7 analyzing post-quantum cryptography algorithms, co-authored peer-reviewed [paper](#) with 5 citations
- Conducted quantum key distribution research implementing BB84 protocol using Qiskit in Python with custom noise models for phonon scattering and charge fluctuations in quantum dot systems

Nanoparticle Synthesis Research Apprentice, US Army

Jun 2023 - Aug 2023

- Engineered autonomous microfluidics robot using C++ and embedded C with multithreaded control of 4-channel valve manifolds and precision syringe pumps (awarded \$3,000 in grant funding)
- Integrated and calibrated UV-VIS spectrophotometer using Ocean Optics SDK, implementing real-time spectral analysis with automated wavelength calibration

PROJECT EXPERIENCE

Shop Story | *Github:* github.com/leahuriarte/shop-story

- Won \$3,000 hackathon prize; selected to productionize app for Shop platform serving over 7 million users
- Built AI-powered analytics mini-app processing user shopping data to generate personalized insights
- Designed Instagram Story-style interface using React and TypeScript for mobile shopping analytics

Sentinel | *Github:* github.com/leahuriarte/sentinel

- Developed autonomous agent to dynamically discover, ingest, and process real-time intelligence to produce briefs
- Engineered multi-modal retrieval pipeline combining vector search, knowledge graphs, and hierarchical summarization

LEADERSHIP

Team Lead, Code the Change for UN High Commission of Refugees

Jan 2025 - Present

- Led 4-person team building React refugee visualization web app, providing technical guidance and code reviews
- Architected interactive mapping application integrating the UNHCR API and processing 24 years of global data
- Engineered demographic visualization system with D3 scaling, Leaflet maps, real-time data processing

SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, HTML/CSS, Google Apps Script

Frameworks & Libraries: PyTorch, TensorFlow, React, Pandas, NumPy, Matplotlib, Seaborn, Qiskit

Cloud & Tools: Google Cloud Platform, Apache Beam, JAX, Git, REST APIs, Docker, Statistical Analysis

AWARDS

Claremont Graduate University Hacks, 1st - Nov 2025 | Shopify Shop Minis Hackathon, Winner - Aug 2025 | Bruin AI Hackathon, Winner - April 2024 | Bloomberg BPuzzled, 2nd - October 2024 | Roy Rocklin Memorial Scholarship - May 2024 | National Merit Scholarship - February 2024 | Treasure Hacks, 2nd - May 2023 | Sonoma Hacks, 1st - Apr 2023